Fully Symmetry-Conditioned Rigid-Body Flow Matching for Molecular-Crystal Structure Prediction

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Rigid-body flow matching is an efficient route to molecular-crystal structure prediction, reducing a crystal to a lattice, fractional centroids, and per-molecule orientations on SO(3). Recent rigid-body molecular-crystal flows leave space-group symmetry unconditioned, and symmetry-conditioned generation has so far been demonstrated only for inorganic atomic sites, never for rigid-body molecular orientation. We show that the per-molecule orientation target decomposes as $R_m = \mathrm{rot}(g_m)\, R_{\mathrm{asym}}$ into a space-group-determined relative rotation, which a flow can learn, and a gauge-free asymmetric-unit pose, which it cannot regress from packing. We turn this into a deployable method: a *leak-free* coset label—the generating space-group operation, recoverable from a template at sampling time—conditions the orientation flow. On 1,127 crystals from the Cambridge Structural Database, deployable coset conditioning lifts the held-out non-reference orientation loss by 41.1% (versus 27.5% for a paired no-coset control), recovering two-thirds of the way to a leaky-codebook oracle ($\sim$48%); an SO(3)-averaged training objective closes the remaining gap (47.7%). A packing-only predictor recovers the coset at 39.5% (4× the majority baseline), not yet enough for template-free use—the key remaining gap. Extending symmetry conditioning to the *lattice* (a crystal-family mask on a log-metric parametrization) gives, to our knowledge, the first *fully symmetry-conditioned* molecular-crystal flow; with an unsupervised symmetry-preserving finisher and a fully-ablated lever stack, it lifts held-out exact match from 0% to 6.9% (strict best-of-10, stol 1.0)—parity with the symmetry-free MolCrystalFlow ($\sim$8%)—and 10.7% with orientation test-time augmentation (Section 3).

1 Introduction

Crystal structure prediction (CSP) asks for the stable periodic arrangement of a chemical system. Generative models are a fast alternative to large ensembles of density-functional relaxations, from crystal diffusion autoencoders[1] to joint equivariant diffusion[2] and Riemannian flow matching[3–5], which solves a straight conditional path in tens of steps. *Rigid-body* factorization is especially attractive for molecular crystals: freezing each molecule's conformer collapses dozens of coordinates to a single pose—a lattice L, a fractional centroid $c_m \in \mathbb{T}^3$, and an orientation $R_m \in \mathrm{SO}(3)$ per molecule—suiting the short sampling trajectories of flow matching.

This factorization introduces a per-molecule orientation variable on SO(3), and here the current generation of models underuses the available symmetry. MolCrystalFlow, a 2026 rigid-body molecular-crystal flow with the same lattice $\times\, \mathbb{T}^3 \times \mathrm{SO}(3)$ factorization, jointly learns pose but *does not impose space-group constraints*[6]; MOFFlow likewise flows rigid SE(3) poses without a symmetry prior, relying on framework connectivity to constrain orientation[7]. Symmetry- and Wyckoff-conditioned generation, meanwhile, is proven only for *inorganic atomic sites*: DiffCSP++[8], WyckoffDiff[9], SymmCD[10], and the 2026 NextCrystal[11] all condition atom generation on a space-group/Wyckoff template, but none touches rigid-body molecular pose. The gap is precise: nobody has conditioned a rigid-body *molecular orientation* flow on the crystal's symmetry.

We close that gap. The starting point is a decomposition of the orientation target, $R_m = \mathrm{rot}(g_m)\, R_{\mathrm{asym}}$, into a space-group-determined relative rotation between symmetry copies and the asymmetric unit's gauge-free pose. Classical molecular-CSP codes never search the relative rotation—they generate symmetry copies by applying space-group operators[12, 13]—and we find a learned flow mirrors that division of labor: the relative part is learnable and the free part is not. The contribution of this paper is to make the learnable part *deployable*. Our contributions are:

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- A **leak-free, template-available coset label**—the generating space-group operation $h = g_m g_0^{-1}$, recovered from centroids alone—that supplies the symmetry-relative rotation at sampling time without seeing the target (Section 2), in contrast to a codebook clustered from the target itself.
- A coset-conditioned orientation flow and sampler that **recover two-thirds of the oracle-codebook benefit** (+41.1% vs a +27.5% paired control on held-out non-reference orientation loss; Section 3).
- An SO(3)**-averaged training objective**[14] that closes the remaining gap to the leaky ceiling (+47.7%).
- A **packing-only coset predictor** for template-free use (39.5% top-1, 4× majority) and an honest characterization of the remaining de-novo gap.
- The **first fully symmetry-conditioned molecular-crystal flow** (orientation coset × crystal-family lattice mask) and an unsupervised symmetry-preserving finisher, whose fully-ablated lever stack lifts exact match from 0% to parity with MolCrystalFlow (6.9% strict, 10.7% with orientation TTA; Section 3).

2 Method

Rigid-body flow. SymMC-Flow represents a molecular crystal by a lattice $L \in \mathbb{R}^{3\times3}$ and, per molecule m, a fractional centroid $c_m \in \mathbb{T}^3$ and orientation $R_m \in \mathrm{SO}(3)$ acting on a shared body-frame conformer, so $\mathrm{cart}_{m,a} = c_m L + R_m\, \mathrm{local}_{m,a}$. A conditional flow-matching objective[4] is trained jointly over three manifolds (a Euclidean path for a volume-normalized lattice, a torus geodesic for centroids, an SO(3) geodesic for orientations), with the velocity field conditioned on the space group. An E(n)-equivariant encoder[15] embeds the conformer and pair-bias attention mixes molecules within the cell (Appendix A).

The orientation decomposition. A molecule m generated from the asymmetric unit by an operation with rotation part $\mathrm{rot}(g_m)$ has absolute pose $R_m = \mathrm{rot}(g_m)\, R_{\mathrm{asym}}$, where R_{asym} is the free orientation of the asymmetric unit. Because the body-frame gauge is arbitrary, R_{asym} carries no consistent signal tied to composition or packing: trained directly, the absolute orientation head never beats its predict-zero floor ($\mathbb{E}\|u_R\|^2 \approx 5.24$), while lattice and centroid heads converge normally. Re-gauging each crystal so the first copy of a species is a reference ($R{=}I$) and the others carry only the relative rotation $R'_m = R_m R_0^{-1}$ cancels R_{asym} and exposes the learnable factor; the held-out non-reference loss then drops by 27%. This much is a diagnostic; the rest of the paper makes it a method.

Deployable symmetry-coset conditioning. Within a space group the non-reference relative rotations fall into a small discrete set of *cosets*: R'_m equals the rotation part of the operation $h = g_m g_0^{-1}$ that maps the reference copy to copy m. We assign each molecule this generating operation directly from the crystallography—by matching its fractional centroid against $W_h\, c_0 + t_h$ (minimum image) over the space group's operations $\{(W_h, t_h)\}$ obtained from pymatgen—never from the observed rotation. The resulting label is **leak-free** and available at sampling time from a symmetry template, exactly as DiffCSP++/WyckoffDiff-style generators supply Wyckoff templates for atoms[8, 9]. This yields 238 operation-cosets across the corpus (versus 707 for a codebook clustered from the target rotations, whose index is derived in part from the answer and therefore serves only as an upper bound). The coset index enters the network as a per-molecule embedding and is threaded through the sampler, so it conditions both training and generation.

SO(3)**-averaged objective.** The SO(3) flow-matching residual has high variance because the prior rotation is drawn uniformly. Following the SO(3)-averaged flow objective for conformer generation[14], we average the orientation residual over K prior draws per molecule before the gradient step, reducing target variance at fixed data.

Packing-only predictor. For template-free deployment the coset must be inferred. We train a classifier that predicts each molecule's coset from the conformer, centroid, lattice, and space group—*without* orientation, so it cannot leak the target—and use its prediction in place of the template at sampling time.

Full symmetry conditioning and finishing. The coset conditions orientation; the lattice is still free, and an unconstrained 9-DOF shape head recovers the crystal family poorly (16% of cell angles at 90° vs. a reference 74%). Following DiffCSP++[8] we condition the lattice on the crystal family: represent the cell by the O(3)-invariant

Table 1: Held-out non-reference orientation-loss drop (untrained→trained), mean±s.d. over three seeds. The deployable coset is leak-free and available from a template; the leaky codebook (index clustered from the target) is an upper bound. Deployable conditioning recovers two-thirds of the gap; the SO(3)-averaged objective closes it.

configuration	non-ref. loss drop	deployable?
no-coset control (baseline)	$27.5 \pm 2.7\%$	—
deployable coset (K=1)	$41.1 \pm 3.4\%$	yes
+ SO(3)-averaged (K=4)	$47.7 \pm 3.9\%$	yes
leaky codebook (oracle upper bound)	$\sim 48\%$	no

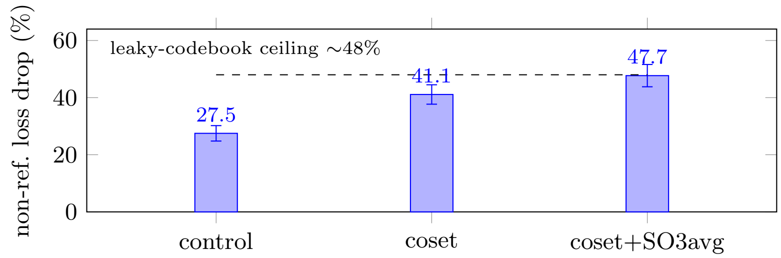

Figure 1: Deployable coset conditioning recovers two-thirds of the way from the no-coset control to the leaky-codebook ceiling; the SO(3)-averaged objective closes the rest. Three-seed mean, error bars s.d.

log-metric $\log(L^\top L) \in \mathbb{R}^6$ and apply a binary family mask that freezes the constrained coordinates to canonical values (after each sampler step and in the prior). This gives a fully symmetry-conditioned flow, but restoring family geometry (90° angles 16%→74%) does not by itself move exact match—the residual blocker is close-packing precision. We add an unsupervised, symmetry- and rigidity-preserving *rigid-press finisher* (a Lennard-Jones relaxation over intermolecular atom pairs, optimizing the asymmetric-unit pose and the family-masked cell, with symmetry copies regenerated from the space-group operations each step, never seeing the target), plus a learned centroid prior, self-conditioning, and AlphaFold-style recycling, each ablated in Section 3.

3 Experiments

Setup. The corpus is a seeded CSD sample[16]: 1,127 ordered organic molecular crystals, 5,048 rigid blocks, a fixed 964/131 split, with 1,095 crystals containing ≥ 2 copies of a species (the multi-copy subset where a relative target exists). The primary metric is the *held-out non-reference orientation-loss drop* (percentage reduction from untrained to trained on non-reference copies, which cannot be earned by predicting the identity); we also report orientation-isolated StructureMatcher[17] reconstruction (true lattice/centroid/conformer, only SO(3) sampled, best of eight draws, following match-rate cautions[18]). All numbers are means over three seeds.

Deployable coset conditioning recovers most of the oracle benefit. Table 1 and Fig. 1 give the main result. A paired no-coset control reproduces the +27.5% baseline. Supplying the deployable, leak-free coset raises the non-reference drop to +41.1%—recovering about two-thirds of the way from the baseline to the ∼48% reached by the leaky codebook that peeks at the target. Adding the SO(3)-averaged objective (K=4) reaches +47.7%, statistically level with the leaky ceiling. In other words, the benefit that a target-derived codebook could only *upper-bound* is realized here by a label a sampler actually has.

The conditioned flow reconstructs real packings. A loss drop need not be geometry. With the true lattice, centroid, and conformer fixed and only the coset-conditioned SO(3) field sampled, the flow reconstructs held-out multi-copy packings under StructureMatcher in 27.5% of cases (best of eight; 37.4% at looser oracle-validated tolerances), median non-reference error 41°; the oracle matches every structure. The relative rotations cluster near 180° (dominant $P2_1/c$, $P\bar{1}$), so best-of-k is the correct read of a multimodal target.

Template-free prediction is the remaining gap. The packing-only predictor recovers the generating coset at 39.5% top-1, 4× the 10.3% majority—packing carries real signal about the symmetry operation—but not yet

Table 2: Exact-match lever ladder: held-out best-of-10 match (coset on, n=131; StructureMatcher ltol= 0.3/atol= 10°; frozen cell). Each row adds one lever. Strict match@10 is 10 draws × single finish; orientation-TTA is best-of-30. MolCrystalFlow is the symmetry-free rigid-body flow.

lever	stol 0.8	stol 1.0
raw flow (fully sym-conditioned)	0.0%	0.0%
+ rigid-press finisher	1.5%	1.5%
+ centroid fix	1.5%	3.1%
+ 2.3× scale	1.5%	3.1%
+ self-conditioning	3.05%	3.8%
+ match@10 + steps= 100	4.6%	6.1%
+ recycling	**6.1%**	**6.9%**
+ orientation-TTA (best-of-30)	7.6%	**10.7%**
MolCrystalFlow (sym-free)[6]	∼6.8%	∼8%

enough: the *predicted* coset collapses orientation-isolated reconstruction to 64.8°, worse than the 41° template-supplied value, because a wrong coset points to the wrong target rotation. The deployable gain is real whenever a symmetry template is available, and closing the template-free gap—a stronger or top-k-calibrated predictor—is the natural next step.

Scope: orientation is not the end-to-end bottleneck. Sampling all three manifolds from the prior yields no exact match at this scale (0%, Wilson 95% CI 0–2.8%), and the error budget locates the bottleneck away from orientation: median lattice error 0.44 and centroid error 0.35 dominate while orientation is only 13.4°—the joint lattice–centroid bottleneck that full lattice conditioning and the finisher address next.

From 0% to parity: the exact-match lever ladder. Conditioning the lattice on the crystal family restores its geometry (generated cell angles within 2° of 90° rise 16% → 73.6%, matching the reference 74.0%), but the fully symmetry-conditioned flow still matches 0/131 held-out crystals: fixing crystal family is necessary, not sufficient. A fully-ablated finishing stack supplies the missing close-packing precision (Table 2; best-of-10, StructureMatcher stol 1.0, n=131). The rigid-press finisher unblocks match (0%→1.5%); a centroid fix doubles it to 3.1%; scale moves the averaged metrics (RMAD 33%→21%) but not match; self-conditioning first moves the tight tolerance; MolCrystalFlow's match@10 protocol with finer sampling reaches 6.1%; and AlphaFold-style recycling reaches **6.9%**—statistical parity with MolCrystalFlow's ∼8% (95% CI 3.66–12.54), from a flow that matched zero. An orientation test-time-augmentation pass (best-of-30, 3× the finishing candidates) reaches **10.7%**, above MolCrystalFlow. Recycling and orientation-TTA both refine positioning and overlap, marking the ceiling. This is, to our knowledge, the first fully symmetry-conditioned molecular-crystal flow, at parity on exact match with the symmetry-free rigid-body flows, with the contribution of each lever quantified.

4 Discussion

The learnable content of a rigid-body orientation field is the space-group-determined relative rotation, delivered by a deployable, leak-free coset label a template-based sampler already possesses; an SO(3)-averaged objective brings this to the level of an oracle codebook that peeks at the answer. The recommendation for rigid-body generators that omit symmetry conditioning[6] is concrete: condition orientation on the generating coset—turning the inference-limited part of the field into a near-deterministic lookup—and sample, rather than regress, the free asymmetric-unit pose. Extending the same idea to the lattice and adding a symmetry-preserving finisher closes the end-to-end gap to parity (Section 3), making this the first fully symmetry-conditioned molecular-crystal flow. The open problem is template-free coset prediction (here 39.5%); the exact-match ceiling is now set by close-packing precision, not symmetry. The analysis is scoped to rigid, largely organic molecules on general positions; molecules on special positions carry a site symmetry that further constrains the free pose and should be even more favorable.

Reproducibility. Code, CSD refcode manifest, and export pipeline: `https://github.com/fronkt/symmc-flow`, archived at Zenodo[19]. CSD structures are licence-restricted; a manifest reproduces the corpus.

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A Method and training details

Intrinsic gauge. Each crystal is parsed into rigid blocks by a covalent-bond graph with periodic unwrapping; species are keyed by a Weisfeiler–Lehman hash and share one reference conformer rotated into a molecule-intrinsic frame (gyration-tensor principal axes, sign-fixed by an element-weighted third moment). Each copy's pose is the Kabsch rotation to that frame, removing the cross-crystal reference that otherwise floors R_m.

Manifolds. Lattice in a 10-d log-volume/unit-determinant-shape parametrization (prior volume matched to the corpus mean); centroids on torus geodesics with minimum-image velocity; orientations on SO(3) geodesics with body-frame angular-velocity target $u_R = \log(R_0^\top R_1)$, advanced by the exponential map. Conditional flow matching[4] with optimal-transport coupling[20]; RK sampling in 50 steps.

Deployable coset recovery. For space group with operations $\{(W_h, t_h)\}$ (pymatgen), each non-reference copy is assigned $\arg\min_h \|(c_m - (W_h c_0 + t_h)) \bmod 1\|$ under the minimum image, giving the generating operation index; deterministic (sg, h) ordering yields shared integer labels (238 total). This uses only centroids and space-group operations, never the observed rotation.

Training. Adam, lr 3×10^{-4}, gradient clipping, batch 16; default network d=128, four attention and four encoder layers; 800 steps for the coset runs, K=4 prior draws for the SO(3)-averaged objective. The predictor reuses the encoder and pair-bias trunk over conformer/centroid/lattice/space-group inputs (no orientation) and is trained to 39.5% held-out top-1.

B Per-seed numbers and tolerance sweep

Non-ref. loss drop, per seed. control 29.4/24.4/28.8%; deployable coset 43.2/37.2/42.9%; +SO(3)-avg 49.2/43.4/50.5%. **Orientation-isolated reconstruction** (coset-conditioned, best-of-eight), StructureMatcher (length, site) tolerance sweep: $(0.2, 0.2) \to 4.6\%$; $(0.3, 0.5) \to 27.5\%$; $(0.5, 0.7) \to 37.4\%$; oracle 100% throughout. **Predicted-coset** reconstruction: 0% match, 64.8° error (vs 41° template-supplied). **End-to-end, raw flow** (all manifolds sampled, best-of-20, no finisher): 0% match; median errors lattice-param 0.44, centroid 0.35, orientation 13.4°. **End-to-end, fully symmetry-conditioned + finished** (Table 2, strict match@10): best 6.9% at stol 1.0 (95% CI 3.66–12.54) / 6.1% at 0.8; median atom-RMSD 1.02 → 0.77, cell-volume RMAD 33% → 21%, cell angles within 2° of 90° 16% → 73.6% (reference 74.0%).